

ALEXANDER AVILEZ

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OBJECTIVE

ECE Master's student at Purdue University with 3+ years of industry experience in firmware and hardware-software security. Currently researching binary verification using hardware performance counters. Seeking to contribute to the Linux Kernel through the LFX Mentorship Program.

EDUCATION

Purdue University	2025 – 2027
Master's in Electrical and Computer Engineering	GPA: 3.76/4.0

National Autonomous University of Mexico (UNAM)	2016 – 2021
Bachelor's in Mechatronics Engineering	

EXPERIENCE

Honeywell	Mexico City, Mexico
Embedded Systems Engineer II	Jan 2022 – Aug 2025

- Developed firmware for automation controllers, managing universal I/Os for sensor acquisition and actuation, and implementing inter-controller communication over building network protocols.
- Developed automated manufacturing test suites to verify PCB integrity, standardizing hardware validation protocols across the production line.
- Integrated ATECC608 secure elements for PKI-based device identity, automating certificate provisioning and firmware signing.

Trustworthy Software Ecosystems Lab – Dr. Santiago Torres-Arias (Purdue)	West Lafayette, IN
Graduate Research Assistant	Sep 2025 – Present

- Built a PMC telemetry pipeline in C to capture hardware performance counter profiles of binary executions, validating that microarchitectural signatures can distinguish software versions for binary-level SBOM verification.

Purdue University	West Lafayette, IN
Graduate Teaching Assistant (ECE 50874: Adv. Software Engineering)	Jan 2026 – Present

- Graded assignments and guided lab decisions for 60 students; delivered weekly feedback on presentations covering software security, formal verification, and validation.

SKILLS

Languages	C, C++, Rust, Python, ARM Assembly, Java, QML
Systems & Frameworks	FreeRTOS, Linux, CMake, Make, Qt
Tools	Git, GDB, Wireshark, Jira, SonarQube, BlackDuck

PROJECTS

OS Kernel Subsystems	Operating Systems, Fall 2025
<i>Virtual Memory & Filesystem Implementation</i>	

- Implemented a filesystem driver with superblock, inode allocation, and full file API, including FIFO and adaptive cache replacement policies.
- Built virtual memory management with demand paging, a coalescing allocator, and copy-on-write fork.

b01lers (Purdue CTF Team) – MITRE eCTF	Spring 2026
<i>Embedded System Security Competition</i>	

- Developed a secure Rust HSM and custom HAL for the TI MSPM0L2228, implementing a protected file system and deploying the design to withstand external exploits while actively compromising rival implementations during the MITRE eCTF.